Curriculum vitae of Dr. Nicolas Mazzocchi

1. General Information

Last name: Mazzocchi
First names: Nicolas Adrien
Title: Doctor of Science
Date of birth: April 29th, 1993
Citizenship: French / EU
Language: French (native), English (professional), Czech (A1)
Email: nicolas.mazzocchi@stuba.sk

Years	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	**total**
Research stay	0	1	0	0	1	0	4	4	6	1	**17**
Peer-review publications international only	**1**	**1**	**2**	**3**	**1**	**2**	**3**	**2**	**4**	**4**	**23**
A*, Q1			1			1				3	**21.74%**
A, Q2	1			2	1		3	1	4	1	**56.52%**
B, Q3		1	1	1				1			**21.74%**
C						1					**4.35%**
Talks with selection committee	3	2	2	1	1	1	2	2	2	0	**16**
Citations WoS	0	0	2	5	2	2	14	8	22	19	**74**
Citations Scopus	0	0	2	4	2	9	18	12	26	27	**100**
Citations Google Scholar	0	1	4	7	8	12	24	29	42	55	**182**

2. Positions

4/2025 – now **Assistant Professor**, *Slovak University of Technology (STU)*, Slovakia
Faculty of Electronic and Computer Science (FEI)

7/2024 – 3/2025 **Researcher**, *Slovak University of Technology (STU)*, Slovakia
Faculty of Electronic and Computer Science (FEI)

1/2022 – 6/2024 **Postdoc**, *Institute of Science and Technology Austria (ISTA)*, Austria
Team of T. A. Henzinger

12/2020 – 12/2021 **Postdoc**, *IMDEA Software Institute*, Spain
Team of P. Ganty

3. Education

2016 – 2020 **PhD in Computer Science**, *Université libre de Bruxelles (ULB)*, Belgium
- 1 week International Winter School, HUJI, 2017
- 2 weeks International Summer School, Marktoberdorf, 2017

2014 – 2016 **Master in Computer Science**, *ENS Paris-Saclay (ENS Cachan)*, France

- 5 months at ULB Begium, ERASMUS+ student, 2016
- 3 months at RWTH Aachen University, Germany, ERASMUS+ student, 2015

2011 – 2014 **Bachelor in Computer Science**, *Aix-Marseille University (AMU)*, France
- 1 month at IMT Alès & Incubator Innov'up, France, 2014
- 1 month at LIS, France, 2013

4. PhD Thesis

Title Contributions to formalisms for the specification and verification of quantitative properties

Supervisors E. Filiot (ULB) and J.-F. Raskin (ULB)

Jury E. Filiot (ULB), Thierry Massart (ULB), J.-F. Raskin (ULB), P.-A. Reynier (AMU), S. Sankaranarayanan (University Colorado Boulder USA)

Graduation October 9th, 2020

5. Grant

2016 **FRIA**, *hosted by ULB*, funded by F.R.S.-FNRS
Doctoral grant of 4 years

6. Research Stay

ISTA, *team of T. A. Henzinger*, Austria
- 2 short visits, 2025
- 2 short visits, 2024
- 2 short visits, 2023
- 2 days invited stay, 2021
- 1 week invited stay, 2018

VUT Brno, *team of O. Lengál*, Czech Republic
- 1 short visit, 2025
- 2 days ERASMUS+ staff mobility, 2024
- 1 short visit, 2023

UMONS Belgium, *team of M. Randour*, Belgium
- 2 weeks invited stay, 2025
- 2 weeks invited stay, 2024

Laboratory i3S, *team of E. Lozes*, France
- 1 week in 2025

Université Louis et Marie Pasteur, *team of I. Jecker*, France
- 1 week ERASMUS+ staff mobility, 2026
- 1 week invited researcher mobility in 2024

University of Liverpool, *team of P. Totzke*, United Kingdom
- 1 month in 2023

7. Responsibilities

International events

Reviewer ATVA, CONCUR, LICS, FSTTCS, FOSSACS, VMCAI, JCSS

PC member CONCUR'26, VMCAI'26 (**both well established international conferences**)

2023 Viennese workshop on Verification, main organizer (35 participants, 28 speakers)

2019 Assistant organizer of RP'19 (80 participants, 34 speakers)

Institutional

10/2025 Appointed EULiST delegate at STU FEI

4/2024 – now STU FEI examination committee member for Bachelor and Master studies

9/2022 – 6/2024 Organization of the team seminars at ISTA

9/2022 – 6/2024 Co-organization of the inter-team (ISTA, TU Wien) seminars at ISTA

5/2021 – 9/2021 Co-organization of the inter-team seminars at IMDEA

1/2017 – 9/2020 Organization of the team seminars at ULB

1/2017 – 9/2020 SVN Administrator of the research team Modelization and Verification at ULB

8. Software

Lead Developer

2024 **QuAK**, developed with N. E. Saraç (ISTA), M. Chalupa (ISTA)
Quantitative automata library for safety-liveness monitoring, checker, and synthesizer. Available on Zenodo and GitHub (peer-reviewed by the TACAS'25 artifact committee)

2022 **FORKLIFT**
Büchi automata Ramsey-based language inclusion checker with antichain heuristics. Available on Zenodo and GitHub (peer-reviewed by the CAV'22 artifact committee)

Participating Developer

2014 **SLAM-Bot**, under the supervision of *J. M. Codol (IMT Alès)*
Simultaneous localization and mapping algorithm for a subaquatic environment. Prototype of a commercialized underwater GPS Software.

2013 **Automata Learner**, under the supervision of *F. Denis (LIS)*
Probabilistic automata spectral learner with randomized singular value decomposition of Hankel matrix, evaluated on PAutomaC challenges.

2013 **Led's Chat**, under the supervision of *P. Niebert (LIS)*

Pure parallel microcontrollers program. Parts of the MP2013 project (Marseille being the European Capital of Culture).

9. Teaching

Teaching Instructor

2023 **Foundation of Model Checking**, at *ISTA*, half-semester, for PhD students
New course introduced to ISTA graduate school

Teaching Assistant

2026 – now **Programming**, *at STU*, full-semester, for undergrads. Teaching of practicals. Managing and grading exams.

2017 – 2020 **Formal Verification**, *at ULB*, full-semester, for graduated students
Preparation and teaching of practicals. Management and grading personalized group projects of 3,4 or 5 students.

2017 – 2020 **Embedded systems**, *at ULB*, full-semester, for graduated students
Preparation and teaching of practicals on computers. Management and grading personalized group projects of 3,4 or 5 students

2017 – 2020 **Fundamental computer science**, *at ULB*, full-semester, for undergrads
Managing and grading a C++ project on SAT solver (MiniSAT)

Supervisor

2025 – 2026 **Supervisor**, *at STU*, Samuel Klement (obtained the Rector's award)
Master thesis on logic for automata with output.

2024 – 2025 **Supervisor**, *at STU*, Alexandra Reviláková (obtained the Rector's award)
Master thesis on automata decomposition.

2022 **Co-supervisor** with T. A. Henzinger, *at ISTA*, Pavol Kebis (now PhD student)
Researcher Internship on operator precedence automata

10. Publications

Particularities of theoretical computer science

Publication strategy In my discipline, conference proceedings are preferred over journals because the review process is considerably faster, allowing the researcher to follow the trends. High-rank conferences (e.g., CAV, ICALP, CONCUR, TACAS) are considered as prestigious publication venues (with acceptance rates around 20-30%). **Submissions to proceedings are peer-reviewed as in journals in other fields** and published in book collections such as Lecture Notes in Computer Science (Springer) or in electronic proceedings such as Leibniz International Proceedings in Informatics (Schloss Dagstuhl), for instance

Extended version Only authors of the best papers in conferences are invited to submit in special issues of computer science journals. Those are again peer reviewed before acceptance or rejection. Extended versions (special issues or not) are required

to include new results that reviewers consider to have a substantial impact despite the existing conference version. The review process for journals is commonly 1-2 years.

Co-authorship: The norm is that **authors are ordered alphabetically** without designation of a main author.

	A*	A	B	C	Q1	Q2	Q3
Invited peer reviewed articles	-	-	-	ISoLA	JCSS	LMCS	IJFCS
Published peer reviewed articles	CAV (x3)	CONCUR (x7), TACAS, FoSSaCS, ICALP, MFCS, FCT	DLT, FSTTCS, RV	-	JCSS	-	-

*Ranking source: conference from Core conference portal, Journal from Resurchify

Articles **invited** in peer-review journals (total 3)

[HMS25] T. A. Henzinger, N. Mazzocchi and N. E. Saraç. Safety and Liveness of Quantitative Properties and Automata. In the Logical Methods in Computer Science (LMCS), year 2025, volume 21, issue 2.

[FMR20] E. Filiot, N. Mazzocchi and J.-F. Raskin. Pattern Logic for Automata with Outputs. In Journal of Foundations of Computer Science, year 2020, volume 31, pages 711–748.

[FMR19] E. Filiot, N. Mazzocchi and J.-F. Raskin. Decidable Weighted Expressions with Presburger Combinators. In Journal of Computer and System Sciences (JCSS), year 2019, volume 106, pages 1–22.

Articles **invited** in conference peer-review proceedings (total 1)

[CHM+24] M. Chalupa, T. A. Henzinger, N. Mazzocchi, and N. E. Saraç. QuAK: Quantitative Automata Kit. In 12th International Symposium on Software Engineering Methodologies Leveraging Applications of Formal Methods, Verification and Validation (ISoLA'24) proceedings, year 2024, Lecture Notes in Computer Science (LNCS) volume 15222, pages 3–20.

Articles published in peer-review journals (total 1)

[JMW25] I. Jecker, N. Mazzocchi and P. Wolf. Decomposing Permutation Automata. In the Journal of Computer and System Sciences (JCSS), year 2026, volume 156, pages 103721.

Articles published in conference peer-review proceedings (total 18)

[AHL+26] O. Alexaj, V. Havlena, O Lengál, Y. Li, N. Mazzocchi. Complementing Emerson-Lei Elevator Automata. In the 37th International Conference on Concurrency Theory (CONCUR'26) proceedings, year 2026, Leibniz International Proceedings in Informatics (LIPIcs) volume 391, pages 8:1–8:22.

[HMS+26] T. Henzinger, N. Mazzocchi, N. E. Saraç, H. Yilmaz. Extending QuAK with Nested Quantitative Automata. In 38th International Conference on Computer

Aided Verification (CAV'26) proceedings, year 2026, Lecture Notes in Computer Science (LNCS) volume 16683, pages 418–432.

[AHH+26] O. Alexaj, V. Havlena, L. Holík, O Lengál, Y. Li, N. Mazzocchi. Kofola 1.0: A Modular Approach to Omega-Regular Complementation and Inclusion Checking. In 38th International Conference on Computer Aided Verification (CAV'26) proceedings, year 2026, Lecture Notes in Computer Science (LNCS) volume 16683, pages 181–197.

[ABM+25] P. Austin, S. Bose, N. Mazzocchi, P. Totzke. Temporal Explorability Games. In the 36th International Conference on Concurrency Theory (CONCUR'25) proceedings, year 2025, Leibniz International Proceedings in Informatics (LIPIcs) volume 348, pages 7:1–7:17.

[HKM+25] T. A. Henzinger, P. Kebis, N. Mazzocchi, and N. E. Saraç. Quantitative Language Automata. In the 36th International Conference on Concurrency Theory (CONCUR'25) proceedings, year 2025, Leibniz International Proceedings in Informatics (LIPIcs) volume 348, pages 21:1–21:24.

[CHM+25] M. Chalupa, T. A. Henzinger, N. Mazzocchi, and N. E. Saraç. Automating the Analysis of Quantitative Automata with QuAK. Accepted for publication in the 31st International Conference Tools and Algorithms for the Construction and Analysis of Systems (TACAS'25).

[HMS24] T. A. Henzinger, N. Mazzocchi and N. E. Saraç. Strategic Dominance: A New Preorder for Nondeterministic Processes. In 35th International Conference on Concurrency Theory (CONCUR'24) proceedings, year 2024, Leibniz International Proceedings in Informatics (LIPIcs) volume 311, pages 29:1–29:20.

[BHM+23] U. Boker, T. A. Henzinger, N. Mazzocchi and N. E. Saraç. Safety and Liveness for Quantitative Automata. In 34th International Conference on Concurrency Theory (CONCUR'23) proceedings, year 2023, Leibniz International Proceedings in Informatics (LIPIcs) volume 279, page 17:1–17:18.

[HMS23] T. A. Henzinger, N. Mazzocchi and N. E. Saraç. Quantitative Safety and Liveness. In 26th International Conference on Foundations of Software Science and Computation Structures (FoSSaCS'23) proceedings, year 2023, Lecture Notes in Computer Science (LNCS) volume 13992, pages 349–370.

[HKM+23] T. A. Henzinger, P. Kebis, N. Mazzocchi and N. E. Saraç. Regular Methods for Operator Precedence Languages. In 50th International Colloquium on Automata, Languages, and Programming (ICALP'23) proceedings, year 2023, Leibniz International Proceedings in Informatics (LIPIcs) volume 261, pages 129:1–129:20.

[DGM22] K. Doveri, P. Ganty and N. Mazzocchi. FORQ-based Language Inclusion Formal Testing. In 34th International Conference on Computer Aided Verification (CAV'22) proceedings, year 2022, Lecture Notes in Computer Science (LNCS) volume 13372, pages 109–129.

[HMS22] T. A. Henzinger, N. Mazzocchi and N. E. Saraç. Abstract Monitors for

Quantitative Specifications. In 22nd International Conference on Runtime Verification (RV'22) proceedings, year 2022, Lecture Notes in Computer Science (LNCS) volume 13498, pages 200-220.

[JMW21] I. Jecker, N. Mazzocchi and P. Wolf. Decomposing Permutation Automata. In 32nd International Conference on Concurrency Theory (CONCUR'21) proceedings, year 2021, Leibniz International Proceedings in Informatics (LIPIcs) volume 203, pages 18:1–18:19.

[FMR+20] E. Filiot, N. Mazzocchi, J.-F. Raskin, S. Sankaranarayanan and A. Trivedi. Weighted Transducers for Robustness Verification. In 31st International Conference on Concurrency Theory (CONCUR'20) proceedings, year 2020, Leibniz International Proceedings in Informatics (LIPIcs) volume 171, pages 17:1–17:21.

[JKM20] I. Jecker, O. Kupferman and N. Mazzocchi. Unary Prime Languages. In 45th International Symposium on Mathematical Foundations of Computer Science (MFCS'20) proceedings, year 2020, Leibniz International Proceedings in Informatics (LIPIcs) volume 170, pages 51:1–51:12.

[FGM19] E. Filiot, S. Guha and N. Mazzocchi. Two-way Parikh Automata. In 39th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS'19) proceedings, year 2019, Leibniz International Proceedings in Informatics (LIPIcs) volume 150, pages 40:1–40:14.

[FMR18] E. Filiot, N. Mazzocchi and J.-F. Raskin. Pattern Logic for Automata with Outputs. In 22nd International Conference on Developments in Language Theory (DLT'18) proceedings, year 2018, Lecture Notes in Computer Science (LNCS) volume 11088, pages 304–317.

[FMR17] E. Filiot, N. Mazzocchi and J.-F. Raskin. Decidable Weighted Expressions with Presburger Combinators. In 21st International Symposium on Fundamentals of Computation Theory (FCT'17) proceedings, year 2017, Lecture Notes in Computer Science (LNCS) volume 10472, pages 243–256.

11. Scientific Communications

Invited Seminars (total 11)

2025 Computer Science, Signals and Systems of Sophia Antipolis (i3S), France

2025 Institute of Science and Technology Austria (ISTA), Austria

2025 Brno University of Technology (VUT Brno), Czech Republic

2024 Université Louis et Marie Pasteur, France

2024 University of Mons (UMONS), Belgium

2024 Brno University of Technology (VUT Brno), Czech Republic

2023 Slovenská technická univerzita v Bratislave (STU), Slovak Republic

2023 University of Liverpool, United Kingdom

2021 Institute of Science and Technology Austria and TU Wien, Austria

2020 Colorado Boulder University (online due to the pandemic), USA

2018 Institute of Science and Technology Austria (ISTA), Austria

Invited Keynote Presentation (total 1)

2025 N. Mazzocchi (based on a joint work with U. Boker, M. Chalupa, T. A. Henzinger, and N. E. Saraç). *Safety and Liveness of Quantitative Properties and Automata.* GT Vérif, 2025, Créteil, France.

Conference talk with selection committee (total 15)

2025 N. Mazzocchi (based on a joint work with M. Chalupa, T. A. Henzinger, and N. E. Saraç.). *Automating the Analysis of Quantitative Automata with QuAK*. At the 31st International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS'25), 2025, Hamilton, Canada.

2024 N. Mazzocchi (based on a joint work with T. A. Henzinger, and N. E. Saraç). *Strategic Dominance: A New Preorder for Nondeterministic Processes*. At the 35th International Conference on Concurrency Theory (CONCUR'24), 2024, Calgary, Canada.

2024 N. Mazzocchi (based on a joint work with U. Boker, T. A. Henzinger, and N. E. Saraç). *Safety and Liveness for Quantitative Automata*. At the 34th International Conference on Concurrency Theory (CONCUR'23), 2023, Antwerp, Belgium.

2023 N. Mazzocchi (based on a joint work with T. A. Henzinger, P. Kebis, and N. E. Saraç). *Regular Methods for Operator Precedence Languages*. At the 50th International Colloquium on Automata, Languages, and Programming (ICALP'23), 2023, Paderborn, Germany.

2024 N. Mazzocchi (based on a joint work with U. Boker, M. Chalupa, T. A. Henzinger, and N. E. Saraç). *Safety and Liveness but Quantitative.* At the 12th Highlights of Logic, Games, and Automata, 2024, Bordeaux, France.

2023 N. Mazzocchi (based on a joint work with T. A. Henzinger, and N. E. Saraç). *Abstract Monitors for Quantitative Specifications*. At the 22nd International Conference on Runtime Verification (RV'22), 2022, Tbilisi, Georgia.

2021 N. Mazzocchi (based on a joint work with I. Jecker, and P. Wolf). *Decomposing Permutation Automata*. At the 32nd International Conference on Concurrency Theory (CONCUR'21), 2021, Virtual Conference.

2020 N. Mazzocchi (based on a joint work with E. Filiot, J.-F. Raskin, S. Sankaranarayanan, and A. Trivedi). *Weighted Transducers for Robustness Verification*. At the 31st International Conference on Concurrency Theory (CONCUR'20), 2020, Virtual Conference.

2019 N. Mazzocchi (based on a joint work with E. Filiot, and S. Guha). *Two-way Parikh automata: Tool in transducer theory*. At the 7th Highlights of Logic, Games, and Automata, 2019, Warsaw, Poland.

2019 N. Mazzocchi (based on a joint work with E. Filiot, and S. Guha). *Two-way Parikh Automata*. At the 39th IARCS Annual Conference on Foundations of Software Technology
and Theoretical Computer Science (FSTTCS'19), 2019, Bombay, India.

2018 N. Mazzocchi (based on a joint work with E. Filiot, and J.-F. Raskin). *Pattern Logic for Automata with Outputs*. At the 6th Highlights of Logic, Games, and Automata, 2018, Berlin, Germany.

2018 N. Mazzocchi (based on a joint work with E. Filiot, and J.-F. Raskin). *Pattern Logic for Automata with Outputs*. At the 22nd International Conference on Developments in Language Theory (DLT'18), 2018, Tokyo, Japan.

2017 N. Mazzocchi (based on a joint work with E. Filiot, and J.-F. Raskin). *Decidable Weighted Expressions with Presburger Combinators*. At the 2nd Winter School in Engineering and Computer Science, 2017, Jerusalem, Israel.

2017 N. Mazzocchi (based on a joint work with E. Filiot, and J.-F. Raskin). *Decidable Weighted Expressions with Presburger Combinators*. At the 5th Highlights of Logic, Games, and Automata, 2017, London, United Kingdom.

2017 N. Mazzocchi (based on a joint work with E. Filiot, and J.-F. Raskin). *Decidable Weighted Expressions with Presburger Combinators*. At the 21st International Symposium on Fundamentals of Computation Theory (FCT'17), 2017, Bordeaux, France.